

Daily Work Report

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Reporting Manager: Aditya Chauhan

Work Summary

Started Week 7 objectives focusing on Scalability and Infrastructure Testing. Successfully established a comprehensive load testing suite using Locust to evaluate system performance under heavy concurrency (100, 500, and 1000 users). Added system instrumentation via psutil to track CPU and Memory metrics in real-time. Conducted deep bottleneck analysis exposing the limitations of the Python GIL during heavy PDF ingestion and embedding generation.

Completed Tasks

- Load Testing Suite: Built Locust scripts (locustfile_query.py and locustfile_mixed.py) to simulate semantic search and heavy document ingestion workloads.
- System Instrumentation: Injected an API middleware to track latency and added /api/v1/health/metrics to expose real-time CPU and Memory usage.
- Automated Execution & Plotting: Wrote Python scripts to automatically execute Locust across multiple concurrency tiers and plot throughput/latency charts using pandas and matplotlib.
- Bottleneck Analysis: Identified that CPU starvation (via the GIL) from sentence-transformers and PyMuPDF causes severe latency degradation at scale.
- Capacity Planning: Authored a comprehensive scaling document recommending horizontal Uvicorn scaling, Redis caching, and a Celery queue for ingestion.

Next Steps / Blockers

Prepare for Week 8 by implementing the infrastructure recommendations, specifically migrating the PDF ingestion pipeline to a background message queue (e.g., Celery) to prevent it from blocking semantic search queries.